

Making Conversational Robots Learn from Children in a Collaborative Museum Exhibition

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ABSTRACT

Teaching Artificial intelligence concepts is incorporated in many formal and informal educational institutions. This demo showcases a novel learning experience of fundamental AI concepts for children in a 30-minute, engaging, hands-on, and playful experience designed for museums and community centers. The experience aims to make children to comprehend that AI systems are built from and by people, and is materialized in a series of activities where the children improve the abilities of three AI-powered conversational robots competing in a quiz game. The demo shows how children can learn about AI by teaching machines in natural language in a way which enables them to grasp intuitively those essential ideas about AI. As part of the formative design process, 109 children interacted and evaluated the prototype.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*.

KEYWORDS

Collaboration exhibition, Artificial Intelligence, Machine Learning

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1 INTRODUCTION

With the advances and increasing relevance of Artificial Intelligence (AI) in computer science and society in general, there has been a renewed interest in having children, especially in the K-12 context, learn basic AI concepts beyond the traditional curriculum of computer science and programming [2, 14]. To understand Artificial Intelligence, children should expect to learn some concepts, discussed in more detail in [5], such as: What is an intelligent machine? How does a machine learn new information? How do human

beings participate in the learning process? How sure are machines about a piece of information they provide?

This demo presents a prototype of a playful, hands-on experience for children to learn about AI and machine learning concepts. The experience was not designed to teach children basic computer programming skills [15] but to provide an introduction to fundamental AI concepts. It focuses on making them learn AI by doing and by thinking about what they do [11, 13]. It is centered on a fundamental task, to teach AI systems to answer questions in natural language, as a way to unveil the mystery of the AI technology [1]. It aims to give awareness of how AI machines work and how influential humans can be in their development process.

It is often said that the best way to learn something is to teach it [11, 13]. In this demo, we showcase a physical space we created where children teach Biology, Chemistry, and Physics to three AI-based conversational robots. In the first deployment, a science museum in Brazil will host this environment. In it, children will discover how standard machine learning algorithms work and how transparent is the process to humans reading the output features. The design of the concept consists of making children (and adults) understand three basic concepts: (1) AI systems use knowledge acquired from human beings; (2) AI systems do not know everything and makes mistakes; and (3) AI systems are corrected and, improved by human beings. We focused on providing what we consider the minimum understanding of AI to enable citizens to make better decisions about AI and society, and to question the black-box nature of AI systems [5]. In practice, children ask questions to the AI-based robots and see that some of them are not able to understand their questions. Then they teach the robots typical ways human pose those questions and experience the robots improve their ability to answer questions because of the knowledge they provided.

AI is already considered a subject to be included in the educational curriculum and competency indicators in many countries [9]. Moreover, previous work also explored practical learning activities in early childhood and elementary schools [17, 18], secondary schools [3]. There is a lack of previous research with middle school students, and this is the main target age of our project. The experience was designed as a hands-on approach towards transparency and explainable AI [16].

At the demo at CSCW, visitors will be able to experience by themselves the whole process we designed for the children, in a compact virtual format. The main message is that AI systems use the knowledge acquired from human beings and can make mistakes [10] but can be corrected and improved by the work of human beings. The main contributions of this demo are:

- (1) To present a novel and collaborative learning experience for children ages 9-13 which was successfully evaluated by 109

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children as part of a formative evaluation for deployment in a Science museum.

- (2) To showcase machine teaching using conversational technologies for non-programmers with a unique, playful activity aimed at bringing awareness, human accountability, and agency for children.

2 THE COLLABORATIVE MUSEUM EXPERIENCE

The learning experience presented in the demo was designed for initial deployment as part of a large exhibit about AI in a science museum in Brazil, to open in 2020.

We co-designed this experience with museum curators and guides to have a prototype concept ready to test with children. The process considered museum curators and staff as co-creators, and together we co-designed several steps of the experience, considering the pace and tailor the process for our target age. In the process, we had the participation of 4 museum curators, two technicians, and 118 museum guides. The process is described in [4] and followed the Design Research methodology and a Human centered and Participatory approach. It consisted in conducting a range of techniques involving empirical work and collaborative co-creation workshops inside the museum. The design research process was composed of: Preliminary Research, Prototyping phase, Assessment phase and, Reflection and Documentation.

The exhibit space resembles the *IBM Watson* matches in 2011, a landmark in AI technology where that computer system defeated two world champions in the TV quiz game *Jeopardy* [8]. In the space (see Figure 1), three talking heads (voice-based chatbots with some basic head movements) compete in a quiz game about science topics the children have just learned in their visit to the museum. During the game, the children see some of the robots being unable to answer some questions. Then the children are invited to improve the robots by working on three touch displays, each connected to one of the talking heads.

Here is a basic description of the experience in the museum:

- (1) *Introduction*: Children are assembled in three groups around the touch displays and the museum guide introduces the term Artificial Intelligence.
- (2) *Initial quiz game*: The guide asks questions to the three robots in a science quiz game and the children see the robots taking turns answering. Each robot has a different personality expressed by animated eyes and mouth and their speech. In each question, one of the talking heads acknowledges not being good at the question topic and visually shows the lower level of confidence to answer the question.
- (3) *Quick tutorial*: The museum guide explains how the robots work, which is by matching the question asked to previously-stored examples, following the intent-action paradigm, often used in commercial dialogue systems such as *IBM Watson Assistant* used in the installation. The guide then tells the children that they can improve the machines' performance by providing more question examples and shows them how to do it.
- (4) *Children teach the machines*: Each of the three groups of children works on one of the touchscreen displays to provide

more examples and improve the performance of a particular robot using a conversational interface, a chatbot system. They insert a new example and test whether the robot can then utter the right answer.

- (5) *Evaluating the teaching process*: The guide repeats the quiz game, asking the questions the kids provided to the robots. The improved performance of the machines is visually shown.
- (6) *Final discussion*: The guide congratulates all and discusses with the children some of the ideas learned.

3 THE DEMO AT CSCW

At CSCW 2021, the demo will provide a richer, more engaging experience than previous showcases we did [6]. The other exhibitions had the aim (1) to evaluate the prototype with potential users; and (2) to demonstrate the research in conversational agent systems supporting multi-user scenarios. The demo at CSCW differs in the language (from Portuguese translated into English) and we tailors the experience to a virtual public, besides showcasing in detail the main interaction of improving the robots. The demo focuses on the machine teaching mode, the fourth step of the experience described before. In this mode, attendees teach questions for one particular bot to learn. Facilitators explain the challenges of teaching machines and how machines connect their knowledge (corpus) to new questions.

The first step is to choose a topic (e.g., Evolution; Human body; Insects). Then the system displays a statement (an answer) to users and suggests users to write a question for the displayed answer in a conversational interface. Second, users write a suitable question, and the bot acknowledges the user, giving feedback on the quality of that question learned. The feedback is displayed by text on the conversational interface, by visual animations on the eyes and mouth of the talking head, by visually displaying the increased system level of confidence score, and by audio using the *text-to-speech* (TTS) technology. Users may teach as many questions to a single topic as they want. After being satisfied with the level of confidence score, users can choose another topic, and follow the same process to teach a new question to the system. At the demo session, the facilitators will be the researchers and we will make available the collaborative training station to the attendees in a virtual manner.

The demo will give the opportunity for attendees to try and experience a machine teaching environment using conversation as a teaching pattern for non-programmers with a unique playful activity and bring awareness and human accountability and agency for children. CSCW attendees will also learn about a state-of-art artificial conversational governance system which allows a natural flow of conversation in a multi-user scenario.

3.1 Envisioned Interaction in the CSCW Demo

Due to physical restrictions imposed by COVID-19, the interaction with the demo will be fully online. However, the participants will see the actual physical installation via a live video stream which will show the response of the actual robots when they are trained. The touch-screen training stations will be replaced by an online user interface which will allow the CSCW virtual participants to



Figure 1: The physical installation in the science museum in Brazil, which will be visible in the live video stream of the demo. From left to right: (1) One robot answering a question and the training stations, (2) The three robots answering a question

interact with the system and see in the live video the robot's response, including the audio, eyes and mouth animation, and head movements.

The following steps present the details of the interaction details of the online participants:

- (1) When attendees enter the virtual exhibit, they can see the physical robots talking with each other via the live video stream. One of the presenters explains the overall demo and one of the robots invites a participant (verbally) to teach an example question to train the model.
- (2) If the participant decides to interact, she receives an external link provided by the presenter. This link redirects the user to a new browser window which shows a conversation interface with a predefined answer and an invitation message of the robot to teach example questions to that specific answer. The interface is shown in Figure 3. All the other attendees see the user interacting with the system.
- (3) The participant uses the online interface and chooses a theme, which triggers a predefined robot answer related to the picked theme. Next, the participant provides a question which matches the robot's answer and sees a reply from the robot. The robot asks then for more example questions based on its confidence level or suggests the attendee to select a new topic. The participant may train the robot with as many questions as she likes.
- (4) After the training, the participant has the opportunity to see the presenter test if the provided questions were learned by the robot on the interface shown in the live video stream. Responses from the robot acknowledges the new question verbally, and the level of confidence is displayed in a monitor below the talking head. The physical reactions of the robots which include the answers (audio), animations (eyes and mouth), and head movements are displayed via the video stream.

A single person or group may enjoy the exhibition. Even though only one person may be typing on the interface, others may participate in suggesting questions and discussing what is best way to teach the robot to provide the right answer. Other attendees of the demo online breakout room see the interaction from any place in front of the screen.

4 TECHNICAL DESCRIPTION

The architecture of the system is based on separated and distributed micro-services which are responsible for each the functional aspect of the exhibit. Starting from the user interface, the participant interacts with the system using their own browsers which contains a web chat interface to train the robots by typing matching questions for a specific robot utterance. This question is routed to a controller micro-service which performs the general synchronization of the other services including an orchestrating platform for multi-agent dialogs [7].

The other main elements of the architecture are a processing sketch to handle the animation and sound, the corpus stored in the cloud-based IBM Watson Assistant service, and the live video stream from the physical installation. The physical elements of the system include a central computer that runs the microservices, robotic heads instrumented with Arduino boards, driving sets of LED matrices for the eyes and mouth, sound speakers, multi-touch computer screens, and monitors to display the robot answer.

5 FINAL REMARKS AND FUTURE WORK

This demo will provide CSCW participants an opportunity to interact with the actual physical installation in the science museum during the demo sessions, selecting questions to ask to robots, and seeing the result in the behavior of the robots answering questions. Researchers, as moderators, will observe who is online in the session and will give instructions on how to proceed during the session presentation.

This demo highlights the importance of including children in the discussion of Artificial Intelligence in a practical, collaborative,



Figure 2: Step-by-step experience of the prototype evaluation at the museum (a) child ask a question to the robots; (b) robots answer the question; (c) display of the level of confidence; and (d) teaching examples questions by conversation.

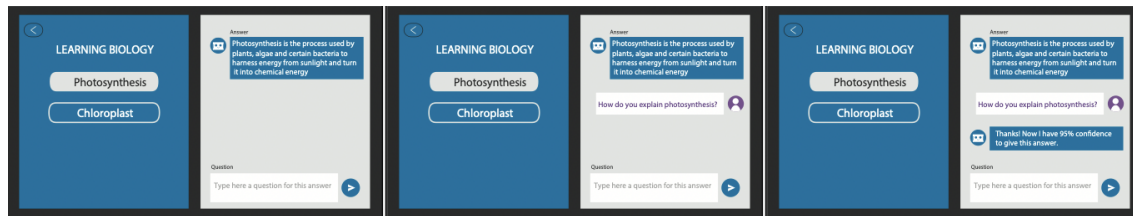


Figure 3: Online interface for training one of the robots. From left to right: (1) The participant chooses a topic and see a robot answer, (2) A new question that match the answer is provided (3) The robot reply to the provided answer

and playful activity. It portrays, in the universe of a children’s exhibition, the actual collaborative process of chatbot development as it is done by professionals [12]. Presenting at CSCW is also an opportunity for the authors to hear feedback and to gain insights before the exhibition goes live in the museum.

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